

P1. Suppose the information content of a packet is the bit pattern 1110 0110 1001 1101 and an even parity scheme is being used. What would the value of the field containing the parity bits be for the case of a two-dimensional parity scheme? Your answer should be such that a minimum-length checksum field is used.

P2. Consider two nodes, A and B, that use the slotted ALOHA protocol to contend for a channel. Suppose node A has more data to transmit than node B, and node A's retransmission probability p_A is greater than node B's retransmission probability, p_B .

a) Provide a formula for node A's average throughput.

- A) $p_A p_B$
 B) $p_A (1 - p_B)$
 C) $p_B (1 - p_A)$
 D) $p_A - p_B$

What is the total efficiency of the protocol with these two nodes?

b) If $p_A = 2p_B$, is node A's average throughput twice as large as that of node B? Why or why not?

- A) No
 B) Yes

If not, how can you choose p_A and p_B to make that happen?

c) In general, suppose there are N nodes, among which node A has retransmission probability $2p$ and all other nodes have retransmission probability p .

i) What is the average throughputs of node A?

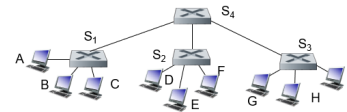
- A) $p(1-p)^{N-1}$
 B) $2p(1-p)^{N-1}$
 C) $2p(1-2p)^{N-1}$
 D) $2p$

ii) What is the average throughputs of any other node?

- A) $p(1-p)^{N-2}(1-2p)$
 B) $p(1-p)^{N-1}(1-2p)$
 C) $2p(1-p)^{N-2}(1-2p)$
 D) $p(1-p)^{N-1}$

P3. Why does collision occur in CSMA, if all nodes perform carrier sensing before transmission?

P4. Consider the self-learning with multi-switch as shown in the figure below. Suppose A sends frame to G, and then G responds to A. Show the switch tables and packet forwarding in S_1, S_2, S_3, S_4 .

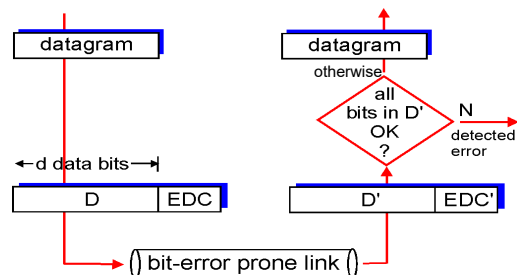


Error detection

EDC= Error Detection and Correction bits (redundancy)
 D = Data protected by error checking, may include header fields

• Error detection not 100% reliable!

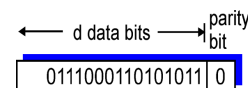
- protocol may miss some errors, but rarely
- larger EDC field yields better detection and correction



Parity checking

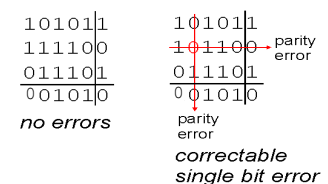
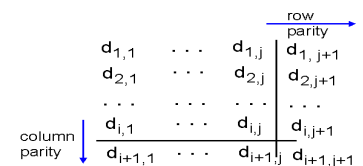
single bit parity:

- detect single bit errors



two-dimensional bit parity:

- detect and correct single bit errors



P1

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1 1 1 0 1
0 1 1 0 0
1 0 0 1 0
1 1 0 1 1
1 1 0 0 0
  
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4x4 → 4 row parities + 4 column parities + 1 corner = 9 parity bits ← fewest

Even Parity scheme: Make the total number of 1s in a group of bits (including the parity bit) even.

Count number of 1s
odd → parity = 1
even → parity = 0

Slotted ALOHA: efficiency

efficiency: long-run fraction of successful slots (many nodes, all with many frames to send)

- ❖ suppose: N nodes with many frames to send, each transmits in slot with probability p
- ❖ prob that given node has success in a slot = $p(1-p)^{N-1}$
- ❖ prob that any one of the N nodes has a success = $Np(1-p)^{N-1}$

- ❖ max efficiency: find p^* that maximizes $Np(1-p)^{N-1}$
- ❖ for many nodes, take limit of $Np^*(1-p^*)^{N-1}$ as N goes to infinity, gives:
max efficiency = $1/e = .37$

at best: channel used for useful transmissions 37% of time!



Link Layer 5-4

P2

- a) A's average throughput is given by $p_A(1-p_B)$.
Total efficiency is $p_A(1-p_B) + p_B(1-p_A)$.

- b) A's throughput is $p_A(1-p_B) = 2p_B(1-p_B) = 2p_B - 2(p_B)^2$.

B's throughput is $p_B(1-p_A) = p_B(1-2p_B) = p_B - 2(p_B)^2$.
Clearly, A's throughput is not twice as large as B's.

- In order to make $p_A(1-p_B) = 2p_B(1-p_A) \Rightarrow p_A - p_A p_B = 2p_B - 2p_A p_B$
 - $p_A = 2p_B - p_A p_B$
 - $p_A + p_A p_B = 2p_B$
 - $p_A(1 + p_B) = 2p_B$
 - $p_A = 2p_B / (1 + p_B)$
- c) A's throughput is $2p(1-p)^{N-1}$, and any other node has throughput $p(1-p)^{N-2}(1-2p)$.

CSMA (carrier sense multiple access)

CSMA: listen before transmit:

if channel sensed idle: transmit entire frame

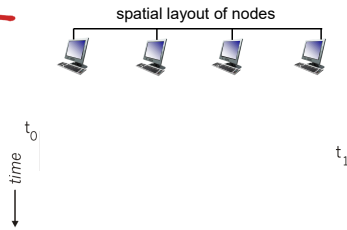
- ❖ if channel sensed busy, defer transmission

- ❖ human analogy: don't interrupt others!

Link Layer 5-6

CSMA collisions

- ❖ collisions *can* still occur: propagation delay means two nodes may not hear each other's transmission
- ❖ collision: entire packet transmission time wasted
 - distance & propagation delay play role in determining collision probability



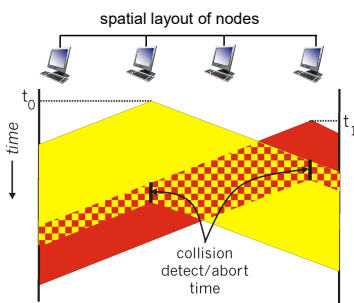
Link Layer 5-7

CSMA/CD (collision detection)

- CSMA/CD:** carrier sensing, deferral as in CSMA
 - collisions *detected* within short time
 - colliding transmissions aborted, reducing channel wastage
- ❖ collision detection:
 - easy in wired LANs: measure signal strengths, compare transmitted, received signals
 - difficult in wireless LANs: received signal strength overwhelmed by local transmission strength
- ❖ human analogy: the polite conversationalist

Link Layer 5-8

CSMA/CD (collision detection)



Link Layer 5-9

P3

- When the bits transmitted by a node A have yet to reach another node B, the latter senses the channel idle and in accordance with the CSMA protocol begins transmitting its frame. A short time later, B's transmission begins to interfere with A's transmission.

P4: Self-learning multi-switch example

- ❖ A sends frame to G: Flooding through S₁, S₄, S₃
- ❖ G responds to A: Selectively send through S₃, S₄, S₁

